

LSTM-AE (uniform MSE + benign-calibrated scoring) — Th = 0.015817, calibrated at FPR(Val) $\leq$ 5%

Rule-Only

Actual	Predicted	
	Normal	Anomaly
Normal	5,674	49
Anomaly	318	1,918

Recall 85.78% F1 91.27%
FPR(Attack) 0.86% FPR(Val) 2.93%

ML-Only

Actual	Predicted	
	Normal	Anomaly
Normal	5,593	130
Anomaly	65	2,171

Recall 97.09% F1 95.70%
FPR(Attack) 2.27% FPR(Val) 2.71%

Hybrid

Actual	Predicted	
	Normal	Anomaly
Normal	5,585	138
Anomaly	35	2,201

Recall 98.43% F1 96.22%
FPR(Attack) 2.41% FPR(Val) 4.97%